



CliQ Merchant API Specifications Integration Document

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1. Overview

1.1 Introduction

This document shows the suggested protocol to the data exchange by specifying the request and response parameters for each API associated with each service.

1.2 Specification

A2A CliQ Merchant Interface is designed to exchange messages using Application Programming Interface (API) which is used for exchanging structured information in its implementation.

These APIs are the integration module that is developed by A2A, and it operates at Merchant side, and its purpose is to provide several web services implemented using REST APIs needed to integrate the functional transactions from and to merchant acceptance system interchangeably, through which, the Merchant system will be able to send the purchase transactions and CliQ system.

The integration system is based on the REST APIs technology over secure http connection. And JSON as the messages format and rely on the HTTP protocols for transmission.

2. Integration Implementation

2.1 REST and JSON

REST (Representational State Transfer) is a web service architectural style. In general, it means that the aspects of the HTTP protocol are used to retrieve or manipulate data from a remote system. As in HTTP, a RESTful API is called with the combination of a URL and a verb. The verb indicates the type of action that is requested, for instance a GET verb means read-only retrieval of information, while the verbs such as POST, PUT and DELETE indicate adding, changing, or removing data.

On the other hand, JSON (JavaScript Object Notation) is a notation style to represent complex object structures in a serialized manner (i.e. transferable over the internet). It has the same role as XML but is much less verbose and therefore faster.

2.2 Security Overview

The REST APIs use two layers of security to ensure two-way authentication of sender and receiver, and to prevent interception of messages and tampering.

The first layer is the TLS that is used for transport security. All calls to the REST API must be done over HTTPS, ensuring end-to-end encryption of the messages and authentication of the system as the recipient of the incoming requests.

2.3 Authentication

Bearer tokens in payment requests is required to access payment APIs. Any party in possession of a bearer token can use it to get access to the associated resources (without demonstrating possession of a cryptographic key). To prevent misuse, bearer tokens need to be protected from disclosure in storage and in transport.

The bearer token is a cryptic string, generated by the server in response to Get Token API request. The client must send this token in the Authorization header when calling requests to payment APIs.

2.4 Requesting REST APIs

To make a call to the provided REST APIs, an HTTP request with the following information should be used:

- The API URL.
- The HTTP verb POST.
- The Content-Type application/json
- A JSON-formatted body containing a valid API request object.

Accordingly, the API will then return a JSON-formatted response body related to the API requested.

3. Header Specifications

A2A Merchant acceptance system is designed to have almost a common request and response schema for most of its services to ease the parsing and manipulation of the JSON data regardless of either building the request or parsing the response.

3.1 HTTP Request Headers

To authenticate the requests coming from the Merchant, a group of values should send as HTTP headers. Moreover, those header values are common for all the REST APIs' requests.

Get Token API Header

Field	Description	Example
CorrelationID	Unique value which is used only once per request and used for further tracking	93ba7b92-7de5-4b46
MerchantID	Merchant ID	521145
UserID	User ID configured and shared by A2A Process Switch system.	TestUser
Password	Generated Password configured and shared by A2A Process Switch system.	TestPassword

Payment Services Header

Field	Description	Example
CorrelationID	Unique value which is used only once per request and used for further tracking	93ba7b92-7de5-4b46
MerchantID	Merchant ID	521145

4. Security Specifications

This specifies security mechanisms employed to secure communication and protect the integrity of messages exchange for the system and client application. The system APIs use two layers of security to ensure two-way authentication of sender and receiver, and to prevent interception of messages and tampering. Where the first layer is the SSL/TLS that is used for transport security. Therefore, all calls to the REST APIs must be done over HTTPS to ensure end-to-end encryption of the messages and authentication of EIB as the recipient of the requests.

4.1 Encryption

All the data transferred between the client Merchant and the system will be encrypted with TLS/SSL, as HTTPS is the protocol used for such communication.

4.2 Access control

- The integration APIs only permit the registered client applications that has a valid certificate to submit the permitted operations to the system.
- Only joined and permitted IP addresses can connect to and use the integration APIs.
- Connection of the client application to the system is authenticated using digital certificates.

4.3 Message integrity, non-repudiation:

- The client application should use the basic authentication of each message sent to the integration APIs. In turns, the integration APIs validates username and password upon receiving the requests.

5. Merchant Services

5.1 Get Token

Service URL

https://xx.xx.xx.xx:0000/A2AMerchantInterface/GetToken

Request

Root Element	Element	Description	Data Type	Required
	SecurityKey	Security Key	String	Yes

Response

Root Element	Element	Description	Data Type
TokenInfo	Token	Token Value	String
	ExpiryDate	Token Expiry Date	DateTime
Result	ErrorCode	Error code (Zero always indicates success response)	Integer
	description	Error description in English	String
	description_ar	Error description in Arabic	String

Error List

HTTP Code	Error Code	Error Description
200	0	Success
400	600	CorrelationID is Mandatory
400	600	MerchantID is Mandatory
400	600	UserID is Mandatory
400	600	Password is Mandatory
401	614	Invalid UserID or Password
500	655	Internal server error
503	650	Service is unavailable

5.1 Merchant Purchase Request

Service URL

https://xx.xx.xx.xx:0000/A2AMerchantInterface/Purchase

Request

Element	Description	Data Type	Required
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MessageTrxID	Unique Transaction ID over the system	String	Mandatory
MerchantID	Merchant ID	String	Mandatory
RAliasType	Customer Wallet alias type (MOBL, ALIAS)	String	Mandatory
RAliasValue	Customer Wallet No entered by the customer	String	Mandatory
Amount	Amount entered by the customer	Decimal	Mandatory
CallBackURL	Merchant Call Back URL endpoint	String	Mandatory

Response

Element	Description	Data Type
errorCode	Error code (Zero always indicates success response)	Integer
description	Error description	String
description_ar	Error description in Arabic	String

Error List

HTTP Code	Error Code	Error Description
200	0	Success
400	304	Error in field
400	304	Missing field - MessageTrxID
400	304	Missing field - MerchantID
400	304	Missing field - RAliasValue
400	304	Missing field - CallBackURL
400	305	Invalid Amount
400	306	Duplicate MessageTrxID
400	307	PSP profile is not defined
500	302	Internal error
200	350	Timeout
200	310	Rejected

5.2 Merchant Payment Inquiry

Service URL

https://xx.xx.xx.xx:0000/A2AMerchantInterface/PaymentInquiry

Request

Element	Description	Data Type	Required
MessageTrxID	Purchase Transaction ID	String	Mandatory

Response

Element	Description	Data Type
errorCode	Error code (Zero always indicates success response)	Integer
description	Error description	String
MSGID	CliQ Message ID	String
MessageTrxID	Purchase Transaction ID	String
RAliasValue	Customer Wallet No entered by the customer	String
Amount	Amount entered by the customer	Decimal
StatusCode	Status Code 0: Success 350: Timeout 300: Rejected 310: Rejected - No response from PSP	String
StatusDescription	Status Description in English	String
StatusDescription_ar	Status Description in Arabic	String

Error List

HTTP Code	Error Code	Error Description
200	0	Success
400	304	Error in field
400	304	Missing field - MessageTrxID
400	304	Missing field - MerchantID
500	302	Internal error

6. Merchant Services Interface

The following section provides an overview of all the APIs that merchants need to integrate on their side in order to receive payment status updates.

The request below should be received on the provided callback URL. The callback URL is the endpoint specified by the merchant to which the payment status updates will be sent by the payment provider. It is essential for the merchant to handle and process these incoming requests appropriately to keep their records updated with the latest payment statuses.

6.1 Payment Status Update

Service URL

https://127.0.0.1:0000/PaymentStatusUpdate {Your full Callback URL}

Request

Element	Description	Data Type	Required
errorCode	Error code (Zero always indicates success response)	Integer	Mandatory
description	Error description	String	Mandatory
MSGID	CliQ Message ID	String	Mandatory
MessageTrxID	Purchase Transaction ID	String	Mandatory
RAliasValue	Customer Wallet No entered by the customer	String	Mandatory
Amount	Amount entered by the customer	Decimal	Conditional
StatusCode	Status Code 0: Success 350: Timeout 300: Rejected 310: Rejected - No response from PSP	String	Conditional
StatusDescription	Status Description in English	String	Conditional
StatusDescription_ar	Status Description in Arabic	String	Conditional

Response

Element	Description	Data Type
errorCode	Error code (Zero always indicates success response)	Integer
description	Error description	String

HTTP Code	Error Code	Error Description
200	0	Success
400	602	Invalid Request Format
202	605	Mandatory parameter is missing
500	655	Internal server error
503	650	Service is unavailable